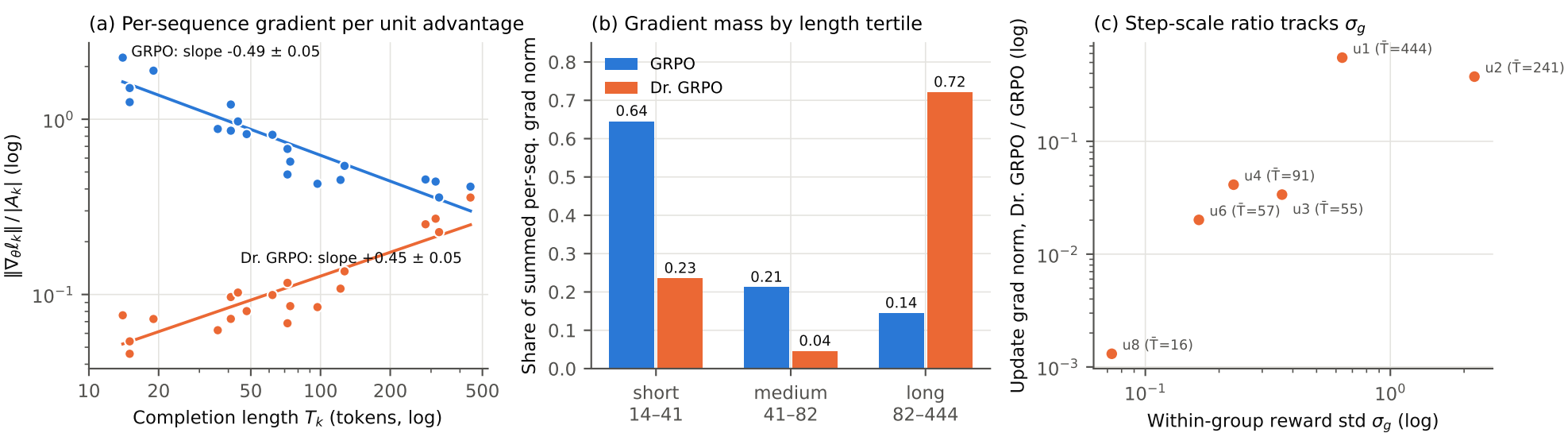
(a) Per-sequence gradient per unit advantage
GRPO: slope -0.49 ± 0.05
Dr. GRPO: slope +0.45 ± 0.05
$\|\nabla_\theta \ell_k\| / |A_k|$ (log)
10^0
10^{-1}
10
20
50
100
200
500
Completion length T_k (tokens, log)
(b) Gradient mass by length tertile
GRPO
Dr. GRPO
Share of summed per-seq. grad norm
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.64
0.23
0.21
0.04
0.14
0.72
short
14–41
medium
41–82
long
82–444
(c) Step-scale ratio tracks σ_g
Update grad norm, Dr. GRPO / GRPO (log)
u1 ($\bar{T}$=444)
u2 ($\bar{T}$=241)
u4 ($\bar{T}$=91)
u3 ($\bar{T}$=55)
u6 ($\bar{T}$=57)
u8 ($\bar{T}$=16)
10^{-1}
10^{-2}
10^{-3}
10^{-1}
10^0
Within-group reward std σ_g (log)